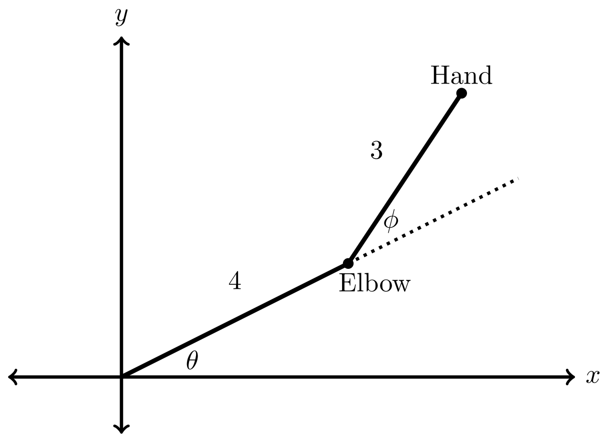
The Mathical Adventures of Robbie the Red
Robot Arms, Ferris Wheels, and Trigonometry

To receive full or partial credit, you must show all work on your own paper.

Robbie the Red often works with robot friends. One of them has a two-link arm given by the diagram below. The "shoulder" is at the origin with angle θ, and the "elbow" is given with angle ϕ. Robbie's friend needs help understanding the location of the hand based on angles provided.

Problem 1. (6 pts) A two-link arm robot is given by the diagram below. From the shoulder at the origin to the elbow, the arm is 4 feet long. From the elbow to the hand, the arm is 3 feet long. Use your knowledge of trig to find the coordinates for the hand's location with the given pairs of angles:

(a) $\theta = \frac{\pi}{3}$ and $\phi = \frac{\pi}{2}$ (b) $\theta = \frac{\pi}{6}$ and $\phi = \frac{\pi}{4}$ (c) $\theta = -\frac{\pi}{4}$ and $\phi = \frac{3\pi}{4}$



Hint: Trig identities may be helpful here.

To celebrate helping his robot friend, Robbie decides to go for a ride on a ferris wheel. In order to relax, Robbie decides to calculate his height after t minutes on the ride.

Problem 2. (6 pts) The ferris wheel has a radius of 20 feet. Robbie begins at the bottom of the circle on a platform that is 5 feet above the ground. It takes 6 minutes to make a complete revolution. Help Robbie find a function in the form below that gives his height after t minutes.

$$h(t) = A\cos(Bt) + D$$

Explore. Robbie chose to use cosine for his height function in the previous exercise. Could he have used sine instead? Repeat the previous exercise with the following function. Assume $A < 0$.

$$h(t) = A\sin(B(t + C)) + D$$

Use the condition $0 \leq C \leq 6$ in this exercise. Would other values for C work without this condition? Use the graph of the function and function transformations to help guide your thoughts.